

AI & ML LAB FILE

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1 Experiment 1: WEKA for Regression

Aim: Explore regression in WEKA.

Procedure:

- Open WEKA Explorer
- Load dataset
- Select Linear Regression
- Run model

Screenshots:

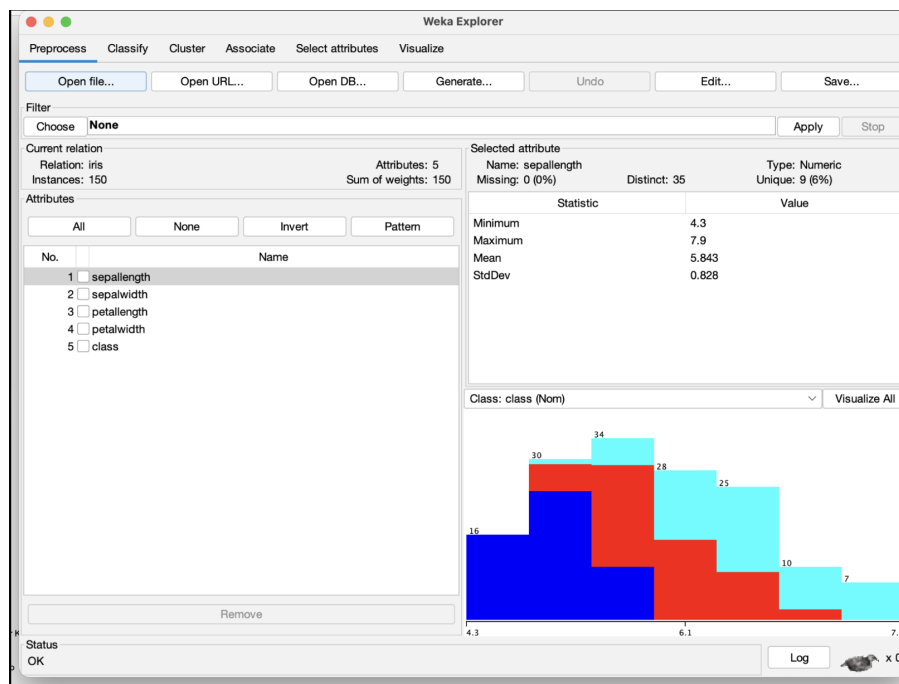


Figure 1: Loading Iris dataset in WEKA

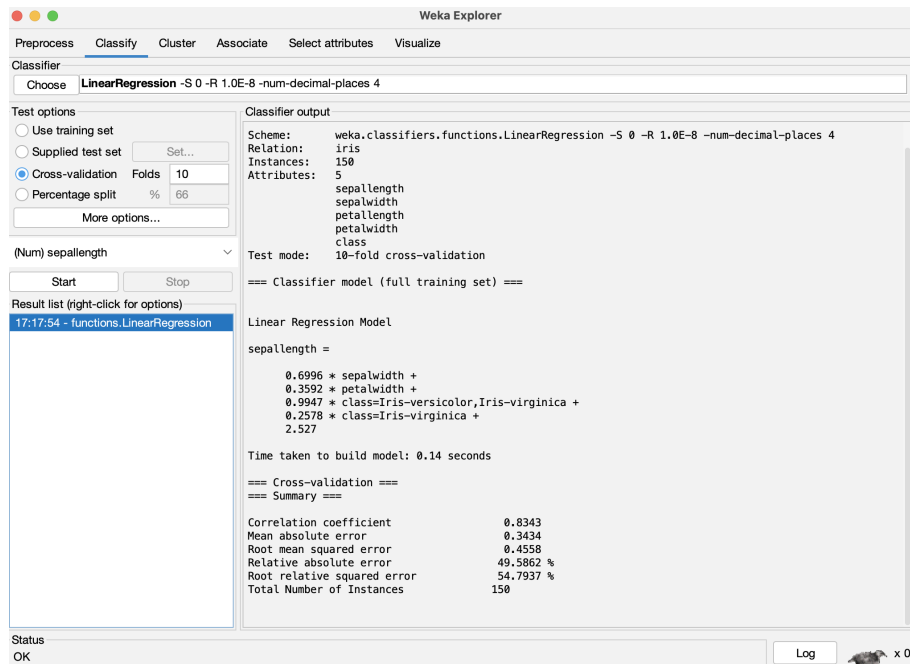


Figure 2: Linear Regression output

Observation:

- Model coefficients observed
- Error rate calculated

Result: Regression successful.

2 Experiment 2: WEKA for Classification

Aim: Perform classification.

Screenshots:

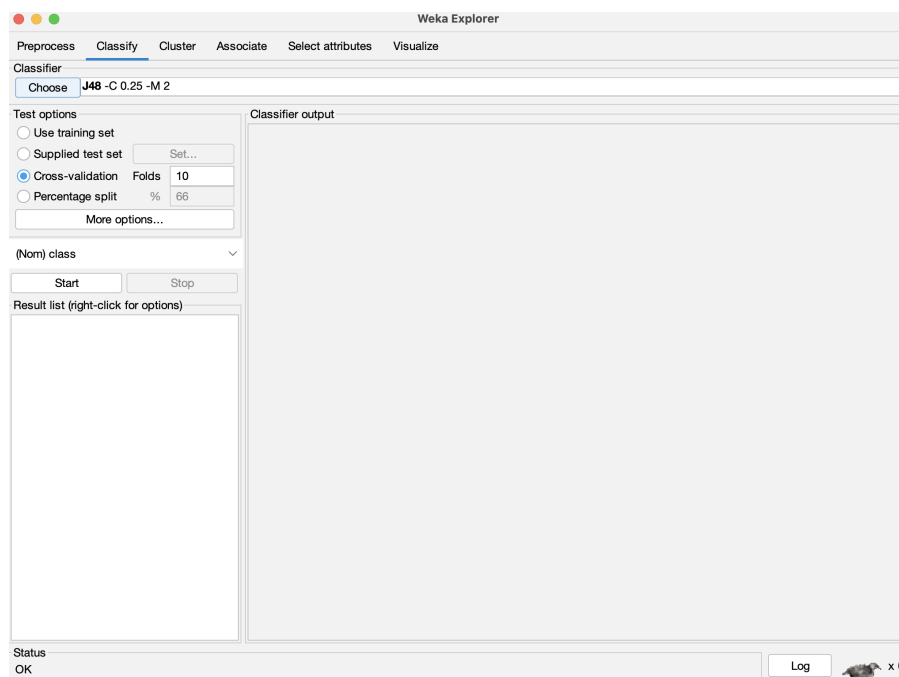


Figure 3: J48 model selection

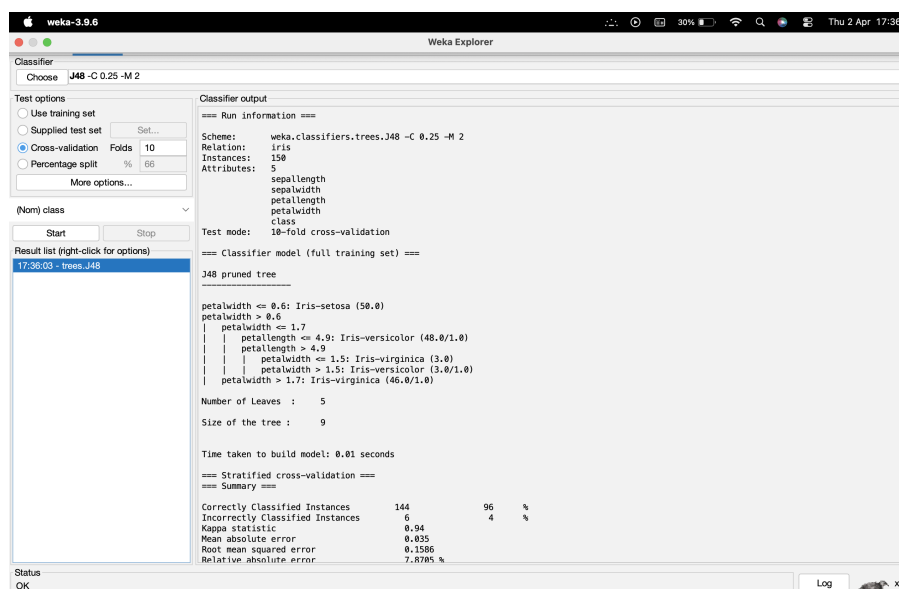


Figure 4: Classification output with confusion matrix

Observation:

- Accuracy percentage observed
- Confusion matrix analyzed

Result: Classification done.

3 Experiment 3: WEKA for Clustering

Aim: Perform clustering.

Screenshots:

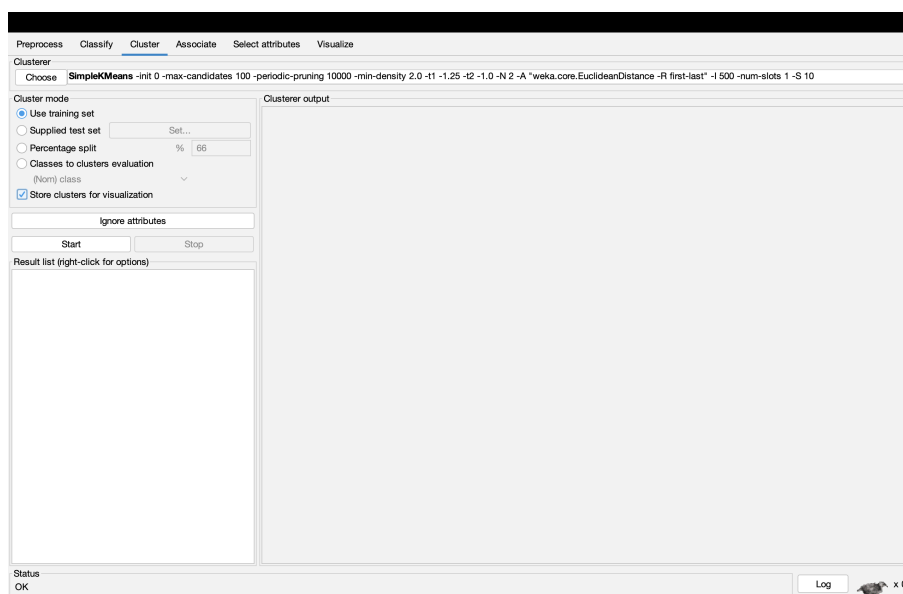


Figure 5: SimpleKMeans setup

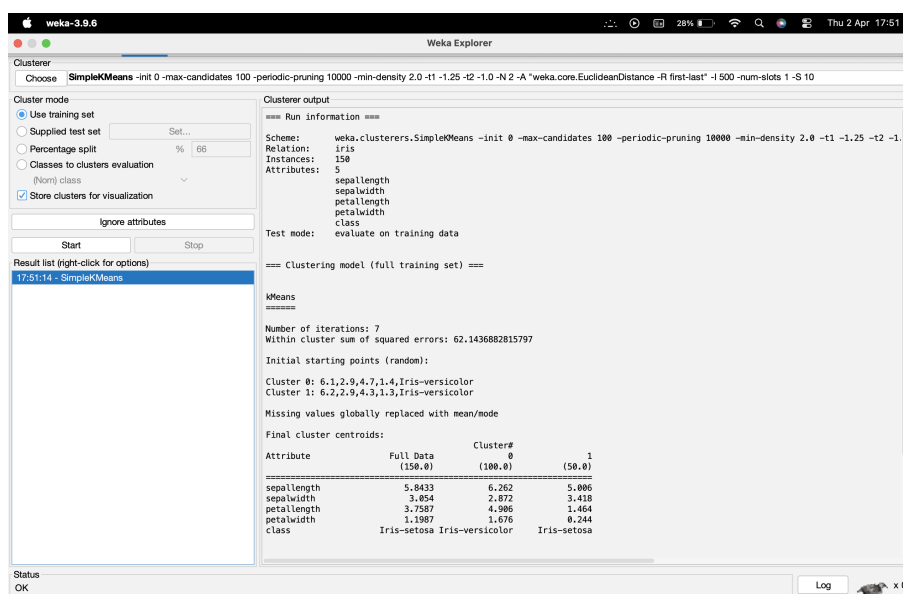


Figure 6: Clustering output

Observation:

- Number of clusters formed

- Data distribution analyzed

Result: Clustering successful.

4 Experiment 4: Data Summary

```
import pandas as pd

df = pd.read_csv('iris.csv')
print(df.shape)
print(df.head())
print(df.describe())
```

Output Table:

Metric	Value	Description
Rows	150	Total records
Columns	5	Total features
Mean	Varies	Average values
Min	Varies	Minimum values
Max	Varies	Maximum values

Observation:

- Dataset has 150 rows
- Statistical summary generated

Result: Dataset analyzed.

5 Experiment 5: Exploratory Data Analysis

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('iris.csv')
print(df.describe())
df.hist()
plt.show()
```

Output Table:

Feature	Mean	Std Dev
Sepal Length	5.8	0.8
Sepal Width	3.0	0.4
Petal Length	3.7	1.7
Petal Width	1.1	0.7

Observation:

- Data distribution visualized

Result: EDA performed successfully.

6 Experiment 6: Missing Values & Outliers

```
import pandas as pd

df = pd.read_csv('iris.csv')

df.fillna(df.mean(numeric_only=True), inplace=True)

Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

df = df[~((df < (Q1 - 1.5*IQR)) | (df > (Q3 + 1.5*IQR))).any(axis=1)]
```

Output:

Missing values replaced

Outliers removed using IQR

Observation:

- Missing values handled
- Outliers removed

Result: Data cleaned successfully.

7 Experiment 7: Scaling & Encoding

```
import pandas as pd
from sklearn.preprocessing import StandardScaler

df = pd.read_csv('iris.csv')

df = pd.get_dummies(df)

scaler = StandardScaler()
df_scaled = scaler.fit_transform(df)
```

Output:

Data encoded and scaled successfully

Observation:

- Data normalized
- Encoding done

Result: Scaling completed.

8 Experiment 8: PCA

```
from sklearn.decomposition import PCA
import pandas as pd

df = pd.read_csv('iris.csv')
df = pd.get_dummies(df)

pca = PCA(n_components=2)
reduced = pca.fit_transform(df)

print(reduced)
```

Output:

```
[[-2.6, 0.5], [-2.5, 0.3], ...]
```

Observation:

- Dimensions reduced

Result: PCA applied successfully.

9 Experiment 9: Imbalanced Data

```
from imblearn.over_sampling import RandomOverSampler
import pandas as pd

df = pd.read_csv('iris.csv')
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

ros = RandomOverSampler()
X_res, y_res = ros.fit_resample(X, y)
```

Output:

Classes balanced successfully

Observation:

- Classes balanced

Result: Imbalance handled.

10 Experiment 10: Visualization

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('iris.csv')

df.plot()
plt.show()

sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```

Output:

Line plots and heatmap displayed

Observation:

- Graphs plotted

Result: Visualization successful.

Conclusion

All experiments were successfully performed.

Thank You